

ML Week 4

Number of participants: 105



1. Lasso and ridge regression

4 correct answers
out of 22 respondents



L1 and L2
penalization both
reduce model
complexity.



36%

8 votes

Lasso and ridge
regression are both
model selection
methods.



36%

8 votes

Lasso and ridge
can produce more
flexible predictive
functions than
usual regression.



45%

10 votes



When feasible,
best subset
selection would be
better than lasso.



41%

9 votes



In general, how could you increase 2. the flexibility of a Ridge regression model?

4 correct answers
out of 23 respondents

Increase the λ
value.



8 votes



Create new
features as
polynomial
functions of the
existing ones.



11 votes



Fit linear model
with least-squares
instead.



10 votes

Use gradient
descent instead of
the closed-form
solution for Ridge.



7 votes



3. Classification theory

0 correct answer
out of 21 respondents



The Bayes classifier is the best possible classifier.



6 votes



We can use the empirical misclassification rate with CV error to choose the k in k NN.



14 votes



With infinitely many observations of (X, Y) , k NN can always recover the Bayes decision boundary.



7 votes



The Bayes classifier predicts perfectly the class of a new test observation.

2 votes



4. In classification:

2 correct answers
out of 19 respondents



Generative models model the distribution of the predictors in each class.



9 votes

Discriminative models can only predict the class but not the class probability.



10 votes

The Bayes decision boundary always has the lowest training error.



9 votes

It is easy to minimize the 0-1 loss numerically.



2 votes